

Supporting Workflows - Audit, Dashboard, Error Handling, and Case Closure Sync

Operational support layer for the GenAI Detection-as-Code MVP: dashboard event collection, global error/dead-letter handling, TheHive closure sync, and shared audit tables.

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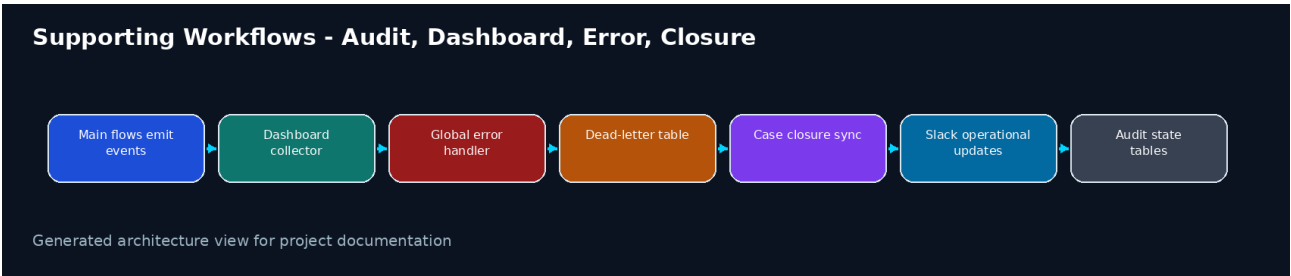
Document Type Supporting workflow project document	Prototype Focus n8n DataTables dashboards error workflow case closure sync
Project Character Portfolio MVP support layer with validation evidence	Design Goal Keep the three main workflows observable, auditable, recoverable, and closed-loop

Why this document matters

- This write-up works both as a portfolio project document and as an implementation-facing prototype note.
- It explains the problem, the workflow design, the SOC value, and the validation evidence that proves the workflow worked in the lab.
- The screenshots are real build/test artifacts collected from n8n, GitHub, Wazuh, Slack, TheHive, and n8n DataTables.

Key outcomes

- Centralizes dashboard metric events from Flow A, Flow B, Flow C, and error handling into flow_soc_dashboard_summary.
- Captures workflow failures through a global Error Trigger and stores them in flow_dead_letter_events while notifying Slack.
- Polls TheHive cases and syncs closed/resolved case outcomes back into Flow C audit tables and Slack.



Architecture view - supporting workflows make the prototype auditable, observable, and closed-loop.

1. Project Summary

The supporting workflows are not the detection logic themselves; they are the operational backbone that makes the prototype easier to monitor, troubleshoot, and prove. They collect dashboard events, capture failures, and synchronize case closures back into the project state.

2. Supporting Workflow Inventory

SOC Dashboard Event Collector	Receives metric events from Flow A/B/C/D and upserts flow_soc_dashboard_summary.
Global Error Dead-Letter Handler	Runs from n8n Error Trigger, normalizes failures, writes flow_dead_letter_events, posts Slack error notifications, and publishes dashboard error metrics.
TheHive Case Closure Sync / Flow D	Polls recently closed Flow C cases, updates flow_c_case_closure_sync and flow_c_audit_events, posts Slack closure notification, and emits dashboard events.
Audit DataTables	Provide a structured evidence layer across CI, deployment, runtime alerting, case promotion, closure sync, and errors.

3. Implementation Evidence

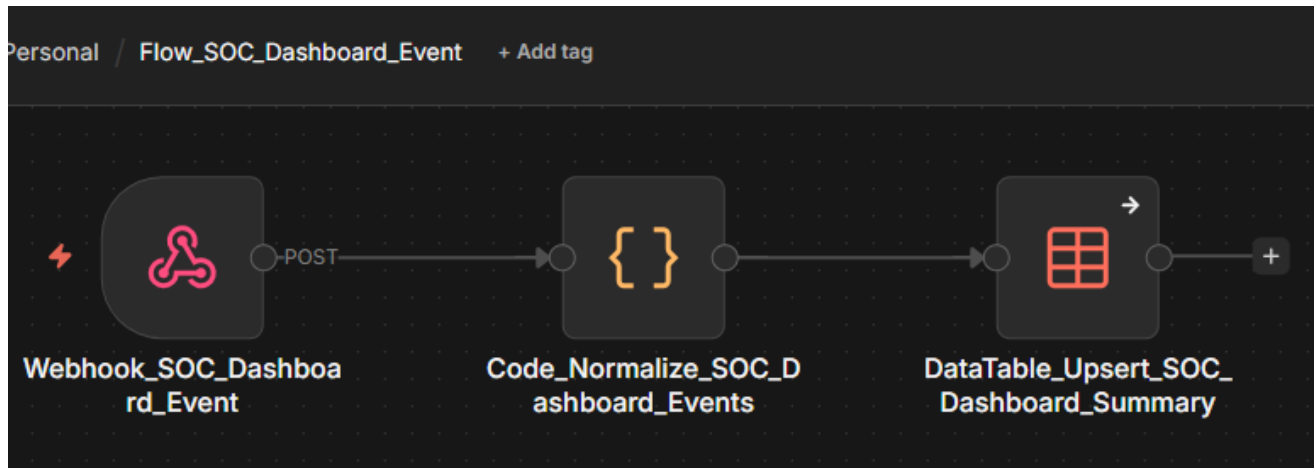


Figure S1 - SOC Dashboard Event Collector workflow: webhook, event normalization, and dashboard summary upsert.



Figure S2 - Global Error Dead-Letter workflow: Error Trigger, normalization, dead-letter DataTable insert, dashboard event, Slack notification.

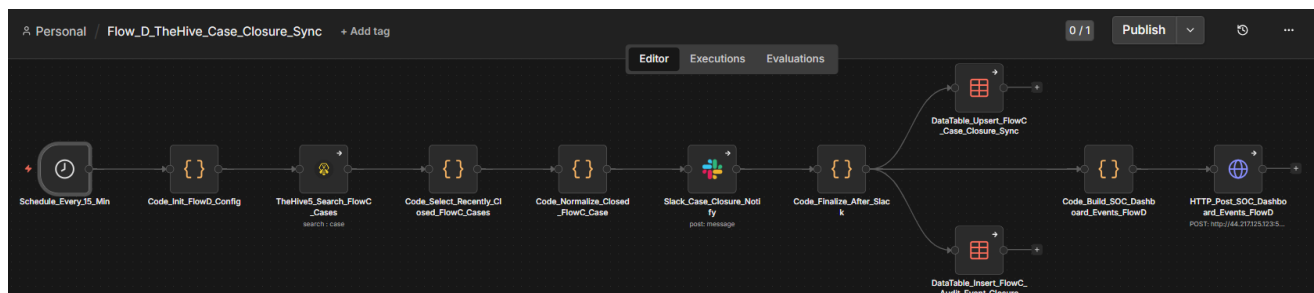


Figure S3 - TheHive Case Closure Sync workflow: schedule/manual trigger, case search, normalization, DataTable update, Slack closure notification.

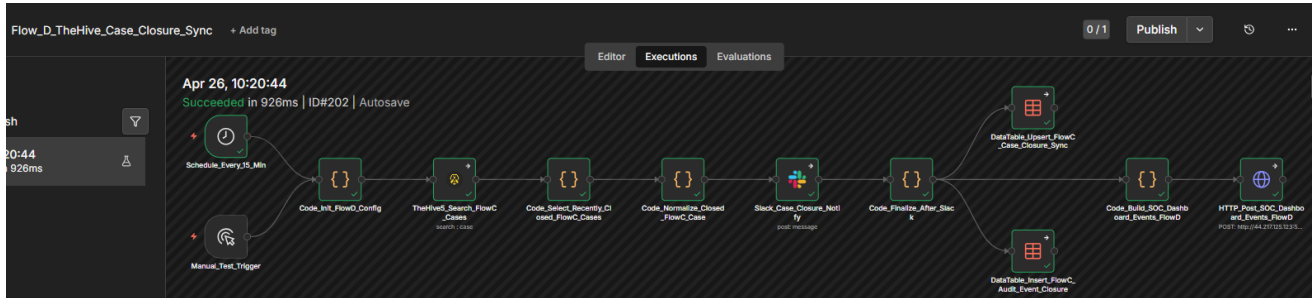


Figure S4 - Extended closure sync workflow showing Slack, audit, and dashboard output path.

```

const cfg = {('Code_Init_FlowC_Config').first(), json;
const lookbackMs = Number(cfg.lookback_minutes || 1440) * 60 * 1000;
const now = Date.now();
const wantedTags = new Set((cfg.flow_c_tags || []).map(x => String(x).toLowerCase()));
const templates = new Set((cfg.flow_c_case_templates || []).map(x => String(x).toLowerCase()));
const closedPatterns = (cfg.closed_status_patterns || []).map(x => String(x).toLowerCase());

function asArray(v) { if(Array.isArray(v)) return v; if(!v) return []; return [v]; }
function lc(v){ return String(v ?? '').trim().toLowerCase(); }
function pickIts(c){
  const candidates = [
    {endDate:c.closedDate,c.closedAt,c.updatedAt,c.updatedAt,c.lastSyncDate,c.startDate,c.date,c_createdAt,c.createdAt};
    for(const v of candidates){
      if(===undefined||v===null||v===) continue;
      if(typeof v==='number') return v>1000000000000 ? v : v*1000;
      const p=Date.parse(v);
      if(!Number.isNaN(p)) return p;
    }
    return null;
  }
  function statusText(c){
    return [
      c.status,c.stage,c.resolutionStatus,c.impactStatus,c.caseStatus,c.workflowStatus,c.summaryStatus].map(lc).join(' ');
  }
}

```

Figure S5 - Code/field mapping evidence for dashboard or error normalization logic.

4. DataTable Evidence

id	T summary_id	window_start	window_end	generated_at	T scope	T flow_name	T metric_name
1	flow:wazuh-genai-ci-pr-26-712e53d2	2026-04-25T05:30:00.000+05:30	2026-04-26T05:29:59.999+05:30	2026-04-25T20:13:48.374+05:30	event	Flow_A_Detection_CI.Validation	ci_runs_total
2	flow:wazuh-genai-ci-pr-26-712e53d2	2026-04-25T05:30:00.000+05:30	2026-04-26T05:29:59.999+05:30	2026-04-25T20:13:48.374+05:30	event	Flow_A_Detection_CI.Validation	ci_fail_count
3	flow:wazuh-genai-ci-pr-26-712e53d2	2026-04-25T05:30:00.000+05:30	2026-04-26T05:29:59.999+05:30	2026-04-25T20:13:48.374+05:30	event	Flow_A_Detection_CI.Validation	relevant_changes
4	flow:wazuh-genai-ci-pr-27-d8af71c27	2026-04-25T05:30:00.000+05:30	2026-04-26T05:29:59.999+05:30	2026-04-25T21:18:27.631+05:30	event	Flow_A_Detection_CI.Validation	ci_runs_total
5	flow:wazuh-genai-ci-pr-27-d8af71c27	2026-04-25T05:30:00.000+05:30	2026-04-26T05:29:59.999+05:30	2026-04-25T21:18:27.631+05:30	event	Flow_A_Detection_CI.Validation	ci_pass_count
6	flow:wazuh-genai-ci-pr-27-d8af71c27	2026-04-25T05:30:00.000+05:30	2026-04-26T05:29:59.999+05:30	2026-04-25T21:18:27.631+05:30	event	Flow_A_Detection_CI.Validation	relevant_changes
7	flowb:flowb-wazuh-genai-ci-pr-26-712e53d2	2026-04-25T05:30:00.000+05:30	2026-04-26T05:29:59.999+05:30	2026-04-25T21:18:27.631+05:30	event	Flow_B_Controlled_Deployment	deployment_runs_total
8	flowc:genai_indirect_injection:req-indir	2026-04-25T05:30:00.000+05:30	2026-04-26T05:29:59.999+05:30	2026-04-26T01:25:53.241+05:30	event	Flow_C_Runtime_GenAI.Triage	genai_alerts_total
9	flowc:genai_prompt_injection:req-direc	2026-04-25T05:30:00.000+05:30	2026-04-26T05:29:59.999+05:30	2026-04-26T01:25:53.241+05:30	event	Flow_C_Runtime_GenAI.Triage	genai_alerts_total
10	flowc:genai_output_handling:req-output	2026-04-25T05:30:00.000+05:30	2026-04-26T05:29:59.999+05:30	2026-04-26T01:25:53.241+05:30	event	Flow_C_Runtime_GenAI.Triage	genai_alerts_total
11	flowc:genai_indirect_injection:req-indir	2026-04-25T05:30:00.000+05:30	2026-04-26T05:29:59.999+05:30	2026-04-26T01:25:53.241+05:30	event	Flow_C_Runtime_GenAI.Triage	thehive_alerts_created
12	flowc:genai_prompt_injection:req-direc	2026-04-25T05:30:00.000+05:30	2026-04-26T05:29:59.999+05:30	2026-04-26T01:25:53.241+05:30	event	Flow_C_Runtime_GenAI.Triage	thehive_alerts_created
13	flowc:genai_output_handling:req-output	2026-04-25T05:30:00.000+05:30	2026-04-26T05:29:59.999+05:30	2026-04-26T01:25:53.241+05:30	event	Flow_C_Runtime_GenAI.Triage	thehive_alerts_created
14	flowc:genai_indirect_injection:req-indir	2026-04-25T05:30:00.000+05:30	2026-04-26T05:29:59.999+05:30	2026-04-26T01:25:53.241+05:30	event	Flow_C_Runtime_GenAI.Triage	cases_promoted
15	flowc:genai_prompt_injection:req-direc	2026-04-25T05:30:00.000+05:30	2026-04-26T05:29:59.999+05:30	2026-04-26T01:25:53.241+05:30	event	Flow_C_Runtime_GenAI.Triage	cases_promoted
16	flowc:genai_output_handling:req-output	2026-04-25T05:30:00.000+05:30	2026-04-26T05:29:59.999+05:30	2026-04-26T01:25:53.241+05:30	event	Flow_C_Runtime_GenAI.Triage	cases_promoted
17	flowd:flowc-closure--4374616:case.cl	2026-04-26T05:30:00.000+05:30	2026-04-27T05:29:59.999+05:30	2026-04-26T10:20:44.909+05:30	event	Flow_D_TheHive_Case_Closure_Sync	case_closures_synced
18	flowd:flowc-closure--4374616:flowc_fa	2026-04-26T05:30:00.000+05:30	2026-04-27T05:29:59.999+05:30	2026-04-26T10:20:44.909+05:30	event	Flow_D_TheHive_Case_Closure_Sync	closed_cases_by_family
19	flowd:flowc-closure--4374616:slack_se	2026-04-26T05:30:00.000+05:30	2026-04-27T05:29:59.999+05:30	2026-04-26T10:20:44.909+05:30	event	Flow_D_TheHive_Case_Closure_Sync	case_closure_slack_notifications

Figure S6 - flow_soc_dashboard_summary table populated with dashboard events from Flow A/Flow B/Flow C.

summary_id	window_start	window_end	generated_at	scope	flow_name	metric_name	metric_value	metric_unit	dimension_key	dimension_value	status_bucket	source_table
flowa:wazuh-genal-ci-pr-26-712e53d2	2026-04-25T05:30:00	2026-04-26T05:29:59.9	2026-04-25T20:13:46	event	Flow_A_Detection_CIValidation	ci_runs_total	1	count	ci_status	skip	skip	flow_a_ci_runs
flowa:wazuh-genal-ci-pr-26-712e53d2	2026-04-25T05:30:00	2026-04-26T05:29:59.9	2026-04-25T20:13:46	event	Flow_A_Detection_CIValidation	ci_fail_count	1	count	ci_status	skip	skip	flow_a_ci_runs
flowa:wazuh-genal-ci-pr-26-712e53d2	2026-04-25T05:30:00	2026-04-26T05:29:59.9	2026-04-25T20:13:46	event	Flow_A_Detection_CIValidation	relevant_changes	0	count	repo	wazuh-genal-ci	skip	flow_a_ci_runs
flowa:wazuh-genal-ci-pr-27-48a77c27	2026-04-25T05:30:00	2026-04-26T05:29:59.9	2026-04-25T21:18:27	event	Flow_A_Detection_CIValidation	ci_runs_total	1	count	ci_status	pass	pass	flow_a_ci_runs
flowa:wazuh-genal-ci-pr-27-48a77c27	2026-04-25T05:30:00	2026-04-26T05:29:59.9	2026-04-25T21:18:27	event	Flow_A_Detection_CIValidation	ci_pass_count	1	count	ci_status	pass	pass	flow_a_ci_runs
flowa:wazuh-genal-ci-pr-27-48a77c27	2026-04-25T05:30:00	2026-04-26T05:29:59.9	2026-04-25T21:18:27	event	Flow_A_Detection_CIValidation	relevant_changes	1	count	repo	wazuh-genal-ci	pass	flow_a_ci_runs
flowb:flowb:wazuh-genal-ci-pr-26-712e	2026-04-25T05:30:00	2026-04-26T05:29:59.9	2026-04-25T21:38:21	event	Flow_B_Controlled_Deployment	deployment_runs_total	1	count	deployment_status	blocked	blocked	flow_b_deployment_runs
flowc:genal:indirect_injectionreq:indiv	2026-04-25T05:30:00	2026-04-26T05:29:59.9	2026-04-26T01:25:53	event	Flow_C_Runtime_GenAl_Triage	genal_alerts_total	1	count	family	genal:indirect_injection	created	flow_c_audit_events
flowc:genal:prompt_injectionreq:direct	2026-04-25T05:30:00	2026-04-26T05:29:59.9	2026-04-26T01:25:53	event	Flow_C_Runtime_GenAl_Triage	genal_alerts_total	1	count	family	genal:prompt_injection	created	flow_c_audit_events
flowc:genal:output_handlingreq:output	2026-04-25T05:30:00	2026-04-26T05:29:59.9	2026-04-26T01:25:53	event	Flow_C_Runtime_GenAl_Triage	genal_alerts_total	1	count	family	genal:output_handling	created	flow_c_audit_events
flowc:genal:indirect_injectionreq:indiv	2026-04-25T05:30:00	2026-04-26T05:29:59.9	2026-04-26T01:25:53	event	Flow_C_Runtime_GenAl_Triage	thehive_alerts_created	1	count	thehive_status	created	created	flow_c_audit_events
flowc:genal:prompt_injectionreq:direct	2026-04-25T05:30:00	2026-04-26T05:29:59.9	2026-04-26T01:25:53	event	Flow_C_Runtime_GenAl_Triage	thehive_alerts_created	1	count	thehive_status	created	created	flow_c_audit_events
flowc:genal:output_handlingreq:output	2026-04-25T05:30:00	2026-04-26T05:29:59.9	2026-04-26T01:25:53	event	Flow_C_Runtime_GenAl_Triage	thehive_alerts_created	1	count	thehive_status	created	created	flow_c_audit_events
flowc:genal:indirect_injectionreq:indiv	2026-04-25T05:30:00	2026-04-26T05:29:59.9	2026-04-26T01:25:53	event	Flow_C_Runtime_GenAl_Triage	cases_promoted	1	count	case_promotion_status	promoted	promoted	flow_c_audit_events
flowc:genal:prompt_injectionreq:direct	2026-04-25T05:30:00	2026-04-26T05:29:59.9	2026-04-26T01:25:53	event	Flow_C_Runtime_GenAl_Triage	cases_promoted	1	count	case_promotion_status	promoted	promoted	flow_c_audit_events
flowc:genal:output_handlingreq:output	2026-04-25T05:30:00	2026-04-26T05:29:59.9	2026-04-26T01:25:53	event	Flow_C_Runtime_GenAl_Triage	cases_promoted	0	count	case_promotion_status	skipped	skipped	flow_c_audit_events
flowd:flowc:closure-4374616:case_cli	2026-04-26T05:30:00	2026-04-27T05:29:59.9	2026-04-26T10:20:44	event	Flow_D_TheHive_Case_Closure_Sync	case_closures_synced	1	count	case_status	TruePositive	closed	flow_c_case_closure_sync
flowd:flowc:closure-4374616:flowc_cli	2026-04-26T05:30:00	2026-04-27T05:29:59.9	2026-04-26T10:20:44	event	Flow_D_TheHive_Case_Closure_Sync	closed_cases_by_family	1	count	family	genal:indirect_injection	closed	flow_c_case_closure_sync
flowd:flowc:closure-4374616:slack_slack	2026-04-26T05:30:00	2026-04-27T05:29:59.9	2026-04-26T10:20:44	event	Flow_D_TheHive_Case_Closure_Sync	case_closure_slack_notific	1	count	slack_status	sent	success	flow_c_case_closure_sync

Figure S7 - flow_soc_dashboard_summary additional columns showing metric value, dimension, status bucket, source table, threshold, breached, and notes.

dead_letter_id	created_at	flow_name	workflow_id	execution_id	node_name	node_type	severity	status	error_message	error_stack	request_id	dupkey	rule_id	flow_stage	original_payload_json	retry_count	retry_after	resolved	owner	resolved_at	notes
No rows																					

Figure S8 - Dead-letter/error table evidence showing captured workflow failure metadata.

5. Why This Layer Matters

- Without dashboard/event collection, the project would only prove one execution at a time; with it, the prototype gains a repeatable KPI evidence layer.
- Without the global error handler, workflow failures can disappear inside n8n executions; with it, failures become Slack notifications and rows in a dead-letter table.
- Without closure sync, TheHive case outcomes remain isolated; with it, closure becomes part of the operational truth in DataTables and Slack.
- This layer demonstrates production-thinking: observability, auditability, and feedback loops.

6. Prototype Limitations and Future Enhancements

- The dashboard table is an event/KPI store, not a fully visual dashboard. Future versions can connect it to Grafana, Metabase, Superset, or n8n reporting pages.
- The global error handler captures workflow execution failures, but node-level continue-on-error paths should still be designed carefully per workflow.
- Case closure sync uses polling; future versions can use TheHive webhooks if available in the deployment.

7. Conclusion

The supporting workflows turn the MVP from a set of automations into an operating model. They make the system observable, auditable, and able to close the loop from detection to case closure.